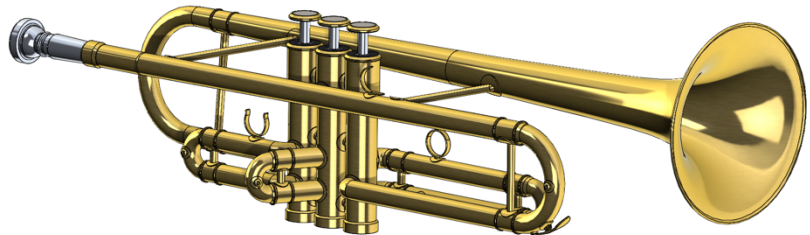
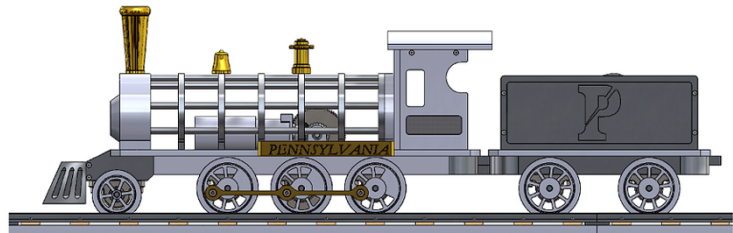


How to CAD Almost Anything!

Summer 2024 – MIT AeroAstro Workshop

A compressed yet rewarding introduction to the parametric design software PTC [Onshape](#), for beginners (no experience at all) and pro-users alike. Come learn how to CAD (computer-aided design) essentially almost anything!



Yes, this could be YOU at the end of the workshop! You'll be equipped with the tools to design cool looking things such as a bicycle, a chess set, a Chinese violin (Er Hu), a model train, a mug and even a Trumpet!

Workshop Details

Subject Title: How to CAD Almost Anything!

Prerequisites: Willingness to have fun and think outside the box!

Enrollment: Unlimited.

Attendance: Participants must attend all sessions.

Meeting Room: [GIS & Data Lab](#) (located on the first floor of the Rotch Library, 7-238).

Meeting Times: Wednesdays 10am-12pm.
There are 8 weekly sessions, starting on 06/05/24 until 08/14/24. The sessions will take place on 06/05, 06/12, 06/18 (Tuesday), 06/26, 07/17, 07/24, 07/31, and 08/14.

Note: The third session, on 06/18/24, falls on a Tuesday due to the Juneteenth holiday!

Instructor: Andy Eskenazi - AeroAstro PhD student (LAE), andyeske@mit.edu.

About: Hailing from the capital of Tango, Steak and Football, Andy is currently a PhD student at MIT AeroAstro trying to make aviation more sustainable. Outside of research, he is passionate about all things mechanical design. At MIT, he has previously taught the “[How to CAD Almost Anything!](#)” series two times, on SolidWorks and Fusion.

Workshop Description

Ever wondered how are objects from our daily lives designed? How can we generate a computer 3D model of a mug, a bottle of Diet Coke, or a Saturn V rocket? What about designing the blades of a jet-engine? A test dummy? How about making an animation of a LEGO house building itself? Or making a realistic render of a bowl of fruit? In this AeroAstro workshop, you will learn the skills to design all of these, and much more!

Split into 8 2-hour long sessions, the first half of each session will be spent learning new PTC [Onshape](#) skills, while the second half will see the application of these new skills through in-class activities. In contrast to traditional mechanical design courses, this workshop places greater emphasis on the design process itself, understanding how we can plan and best leverage our available tools to arrive to our desired result. Thus, the sessions are less about following the instructions on an engineering drawing, but about independent thinking and strategizing, reverse engineering an object into a 3D model.

Workshop Schedule

#	Month	Day	Date	Outline and Objectives
1	June	W	06/05	Session 1: Introduction to Onshape. <u>Objective:</u> In this session, we'll get ourselves acquainted with the Onshape workspace, and start learning some of the most used tools. Session 1's goals include: • Creating sketches (using basic shapes, construction lines, smart-dimensioning, sketch relationships) and understanding planes. • Understanding what it means for a sketch to be fully defined. • Locating and using the different elementary feature commands (boss extrude, boss cut, fillet, chamfer). • Editing sketches and features after creating them. • Coloring parts and changing material properties. <u>Session activity:</u> Using the tools learned on Session 1, we'll design two objects, namely: • A cake • A Casio MS-80B 8-digit calculator.
2	June	W	06/12	Session 2: Splines, sketch pictures and logos! <u>Objective:</u> In this session, we'll continue exploring some of the most powerful Onshape tools. Session 2's goals include:

				• Learning how to use the spline tool. • Learning how to add a picture and sketch on it. <u>Session activity:</u> Using the tools learned on Session 2, we'll design two objects, namely: • A United Airlines safety card (only the front side). • A keychain of your favorite logo.
3	June	T	06/18	Session 3: All about symmetry, patterns and planes! <u>Objective:</u> In this session, we'll focus our attention to symmetry, patterns and planes, and how we can leverage certain tools to simplify the design process. Session 3's goals include: • Understanding how to create a sketch for a revolve. • Learning how to make use of the mirroring and circular patterns tools, both as a sketch and as a feature. • Learning how to create planes, at different angles. <u>Session activity:</u> Using the tools learned on Session 3, we'll design two objects, namely: • A fidget spinner! • A medieval (toy) castle.
4	June	W	06/26	Session 4: Sweeping through a marble run! <u>Objective:</u> In this session, we'll explore two very powerful Onshape tools, loft and sweep. These tools allow us to create complicated looking geometries, like the body of a 747 or the surface of an apple. Session 4's goals include: • Learning how to use the lofting and sweep command. • Continuing to master previously explored tools, such as revolve, linear/circular patterns and plane creation. <u>Session activity:</u> Using the tools learned on Session 4, we'll design two objects, namely: • A simple marble run, with rails! • A banana, in honor of the banana lounge (using various plane cuts, splines and lofts).
5	July	W	07/17	Session 5: Let's have a 3D-modeled Coke in space! <u>Objective:</u> In this session, we'll revise all of our previously learned commands (+ the useful wrap), to make sure we know how to properly use all of them. Session 5's goals include: • Revising some of the previously learned commands, including loft, revolve, sweep, plane creations, patterns, filleting, and material properties. • Learning how to employ the wrap command (for engravings). • Making renders. <u>Session activity:</u> Using the tools learned on Session 5, we'll design two objects, namely:

				• A Diet Coke 8oz glass bottle (with labels included). • A realistic-looking Ariane 5 rocket.
6	July	W	07/24	Session 6: Living in a world made from plastic bricks! <u>Objective:</u> In this session, we'll move towards one of the most powerful features within CAD parametric software, which is that of making assemblies! Session 6's goals include: • Learning how to make an assembly of multiple parts. • Learning (time-permitting) how to make an exploded view of an assembly and subsequently animating it. <u>Session activity:</u> Using the tools learned on Session 6, we'll design a set of objects, namely: • A standard 2x4 LEGO block (as well as 2x3, 2x2 and 2x1). • A standard 2x4 LEGO plate. • A simple LEGO boat assembly.
7	July	W	07/31	Session 7: Variable tables and equations! <u>Objective:</u> In this session, we'll investigate two often underappreciated yet extremely useful tools, variable tables and equations. These tools allow the user to create various configurations of the same model, depending on specific needs. Session 7's goals include: • Changing names of sketches and features under the feature tree. • Learning how to create equations and incorporate them into a design table. • Creating configurations of the same model. <u>Session activity:</u> Using the tools learned on Session 7, we'll design: • A kart from MarioKart, using multiple configurations of kart and wheel sizes, as well as available colors.
8	Aug	W	08/14	Session 8: It is not engineering without drawings! <u>Objective:</u> In this last session, we'll exploit one of Onshape's built-in best features, its drawing-creation workspace. In essence, after creating a part or assembly model, it is possible to create an engineering drawing of it. Session 8's goals include: • Learning how to create an engineering drawing of a part and assembly (including exploded views). • Exploring the best-practices of drawing creation, including different formats and information to be conveyed (such as tolerancing, material properties). <u>Session activity:</u> Using the tools learned on Session 8, we'll design a variety of items, including: • A laser-cuttable wooden airplane model. • A set of engineering drawings of the plane assembly and each of its constituent parts.